

AeroEdge-X

Tata Technologies InnoVent-27 Stage 02 Pre-Read

Implementation Audit & Technical Figures

AeroEdge-X Pre-Read Technical Figures — Implementation Status

> **Document Purpose**: Maps each generated figure to its verified implementation
> status, the source files that provide evidence, and the classification
> methodology used.
>
> **Source of Truth**: Current repository at `d:\Tata Innovent\Tata\_Innovent-main`
> **Generated**: 2026-10-02
> **Generator Script**: `docs/pre-read/figures/generate\_figures.py`

Status Classification

Label	Color Code	Meaning
VERIFIED	#12C6B3` (Teal)	Concrete code exists and runs in the current repository
PARTIAL	#FF9C00` (Amber)	Some implementation exists but the feature is not complete
PLANNED	#FF9C00` (Amber outline)	Designed / documented but no running code in repo
TARGET	#5F6673` (Grey)	Future milestone; no code or configuration present

Figure Index & Evidence

Figure 13 — End-to-End Technician Workflow

- **File**: `figure\_13\_technician\_workflow.png`
- **Overall Status**: ■ VERIFIED
- **Evidence**:
 - `frontend/src/` — React UI with inspection workflow pages
 - `backend/agents/vision\_agent.py` — YOLOv11 defect detection
 - `backend/agents/rag\_agent.py` — ChromaDB retrieval + maintenance docs
 - `backend/agents/reasoning\_agent.py` — Phi-3 Mini via Ollama

- `backend/services/digital\_twin.py` — SQLite persistence
- `backend/services/report\_generator.py` — PDF report generation

Figure 14 — AI Inspection Pipeline

- **File:** `figure\_14\_ai\_inspection\_pipeline.png`
- **Overall Status:** ■ VERIFIED
- **Evidence:**
 - `backend/agents/vision\_agent.py` — OpenCV preprocessing → YOLOv11 detection → bounding box + confidence + class
 - `backend/agents/rag\_agent.py` — ChromaDB semantic search with `all-MiniLM-L6-v2` embeddings, `PyMuPDFLoader` document loading
 - `backend/agents/reasoning\_agent.py` — Phi-3 Mini reasoning via Ollama local API
 - No TensorRT / ONNX optimization code found → marked as PLANNED on diagram

Figure 15 — Current System Architecture

- **File:** `figure\_15\_current\_system\_architecture.png`
- **Overall Status:** ■ VERIFIED (all layers)
- **Evidence:**
 - Electron shell: `main.js`, `package.json` (`electron`, `electron-builder`)
 - React UI: `frontend/src/`
 - Flask REST API: `backend/app.py`
 - Agent orchestration: `backend/agents/`
 - AI/Knowledge: OpenCV, YOLOv11 (`ultralytics`), ChromaDB, `sentence-transformers`, Ollama
 - Data: SQLite (`digital\_twin.py`), local images, PDF reports (`report\_generator.py`)

Figure 16 — Edge + Cloud Evolution

- **File:** `figure\_16\_edge\_cloud\_evolution.png`
- **Overall Status:** Mixed
- **Column 1 (Current Verified):** ■ All components verified per Figure 15
- **Column 2 (Next Implementation):** ■■ PLANNED
- No Jetson-specific libraries (`jetson-inference`, `JetPack`) found

- No `VideoCapture` / industrial camera code found
- No TensorRT scripts found
- No sync queue implementation found
- **Column 3 (Target Cloud):** ■ ■ TARGET
- No `boto3`, no AWS SDK, no API Gateway config, no DynamoDB/S3 code

Figure 17 — Offline-First Synchronization

- **File:** `figure\_17\_offline\_sync\_workflow.png`
- **Overall Status:** Mixed
- **Verified (left column):**
 - SQLite local save: `digital\_twin.py`
 - Local artifacts (images + reports): file-based storage
- **Planned (center/right):**
 - No sync record creation code
 - No offline queue / retry logic
 - No AWS S3/DynamoDB upload code
- `rag\_agent.py`: `HF\_HUB\_OFFLINE=1` confirms offline-first design intent

Figure 18 — Jetson Edge Deployment

- **File:** `figure\_18\_jetson\_edge\_deployment.png`
- **Overall Status:** ■ ■ PLANNED / TARGET
- **Evidence:**
 - No `jetson-inference` or `JetPack` imports
 - No TensorRT / ONNX conversion scripts
 - No `tegrastats` or GPU monitoring
 - Architecture is designed for portability (Flask + agents pattern)

Figure 19 — AWS Cloud Architecture

- **File:** `figure\_19\_aws\_architecture.png`

- **Overall Status:** ■■ TARGET
- **Evidence:**
- No `boto3` in `requirements.txt` or any `.py` file
- No Lambda, API Gateway, S3, DynamoDB, or CloudWatch code
- Diagram shows planned target architecture with all components marked as TARGET

Figure 20 — Inspection Data Flow

- **File:** `figure\_20\_inspection\_data\_flow.png`
- **Overall Status:** ■ VERIFIED (core flow)
- **Evidence:**
- Image upload → `vision\_agent.py` (YOLOv11) → `rag\_agent.py` (ChromaDB) → `reasoning\_agent.py` (Phi-3) → UI display → `report\_generator.py` (PDF)
- Cloud sync portion marked as PLANNED

Figure 21 — Evidence Traceability

- **File:** `figure\_21\_evidence\_traceability.png`
- **Overall Status:** ■ VERIFIED
- **Evidence:**
- Visual evidence: image + YOLOv11 bounding box + confidence
- Maintenance evidence: ChromaDB retrieval with source doc + page number
- Generated guidance: Phi-3 Mini reasoning grounded in retrieved context
- Human review: technician review step in UI
- Recorded output: SQLite record + PDF report

Figure 22 — Current Status & Roadmap

- **File:** `figure\_22\_current\_status\_roadmap.png`
- **Overall Status:** N/A (status overview diagram)
- **Classification:**
- M1 (Core AI POC): COMPLETED — all core AI agents verified
- M2 (Architecture & UI): COMPLETED — Electron + React + Flask verified

- M3 (Jetson + Camera + AWS): NEXT — no code evidence
- M4 (Prototype Validation): REMAINING — pipeline integration pending
- M5 (Stage 3 Target): JAN 2027
- M6 (Future Scope): BEYOND S3

Figure 23 — Future Evolution

- **File**: `figure\_23\_future\_evolution.png`
- **Overall Status**: N/A (vision/roadmap diagram)
- All items shown are future targets with no current code evidence

Figure 24 — Prototype Validation Plan

- **File**: `figure\_24\_validation\_matrix.png`
- **Overall Status**: Mixed (matrix format)
- **Verified rows**:
 - Offline Inspection: `rag\_agent.py` HF_HUB_OFFLINE=1
 - Defect Detection: `vision\_agent.py` YOLOv11 via ultralytics
 - RAG Retrieval: `rag\_agent.py` ChromaDB + PyMuPDFLoader
 - Local Reasoning: `reasoning\_agent.py` Ollama Phi-3 Mini
 - Persistence: `digital\_twin.py` SQLite log_inspection()
 - Reporting: `report\_generator.py` ReportLab
- **Planned rows**:
 - Jetson: No JetPack / TensorRT scripts
 - Camera: Upload works, no VideoCapture
 - Synchronization: No sync queue or AWS code
 - Recovery: No offline queue roadmap code
 - Performance: Benchmarking planned for Stage 3

Figure 25 — Architecture Evolution

- **File**: `figure\_25\_architecture\_evolution.png`

- **Overall Status**: N/A (evolution timeline diagram)
- Phase 1 (Current): All components verified
- Phase 2 (Next): Jetson + edge — PLANNED
- Phase 3 (Target): Full cloud integration — TARGET

Verification Methodology

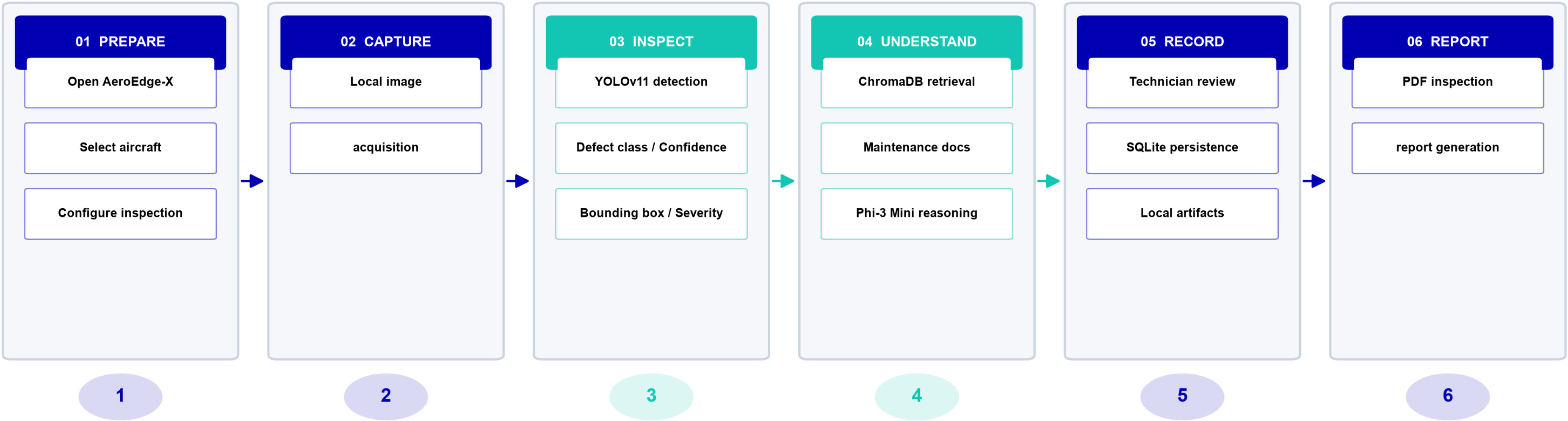
1. **Grep search** for key imports: `boto3`, `jetson`, `TensorRT`, `VideoCapture`, `onnx`
2. **File inspection** of all `backend/agents/\*.py`, `backend/services/\*.py`, `backend/config.py`
3. **Package audit** of `requirements.txt` and `package.json`
4. **No assumptions** from README, PPT, Figma, or comments

File Listing

#	Filename	Dimensions	Status
13	figure_13_techician_workflow.png	3810 × 2169	VERIFIED
14	figure_14_ai_inspection_pipeline.png	3810 × 2169	VERIFIED
15	figure_15_current_system_architecture.png	3810 × 2169	VERIFIED
16	figure_16_edge_cloud_evolution.png	3810 × 2169	MIXED
17	figure_17_offline_sync_workflow.png	3810 × 2169	MIXED
18	figure_18_jetson_edge_deployment.png	3810 × 2169	PLANNED
19	figure_19_aws_architecture.png	3810 × 2169	TARGET
20	figure_20_inspection_data_flow.png	3810 × 2169	VERIFIED
21	figure_21_evidence_traceability.png	3810 × 2169	VERIFIED
22	figure_22_current_status_roadmap.png	3810 × 2169	N/A
23	figure_23_future_evolution.png	5890 × 2169	N/A
24	figure_24_validation_matrix.png	3810 × 2400	MIXED

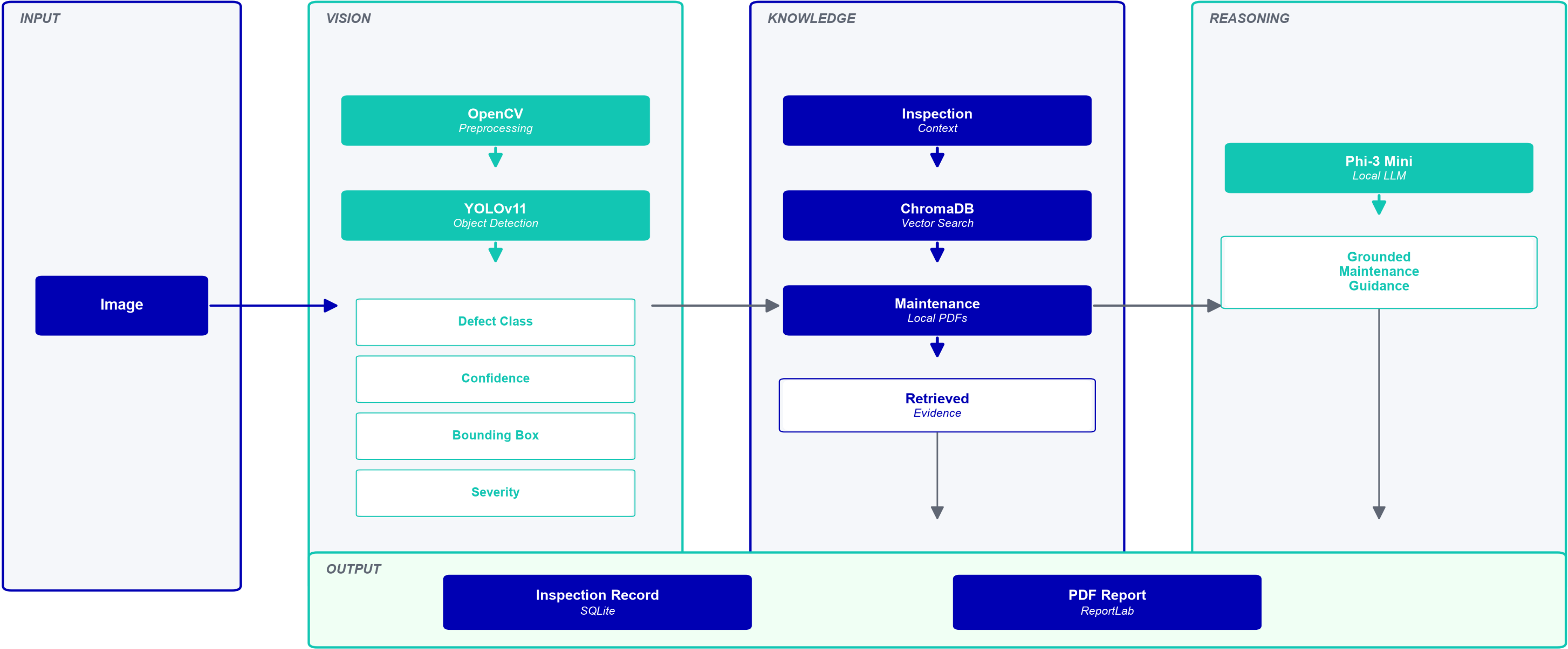
AeroEdge-X End-to-End Technician Workflow

CURRENT VERIFIED WORKFLOW



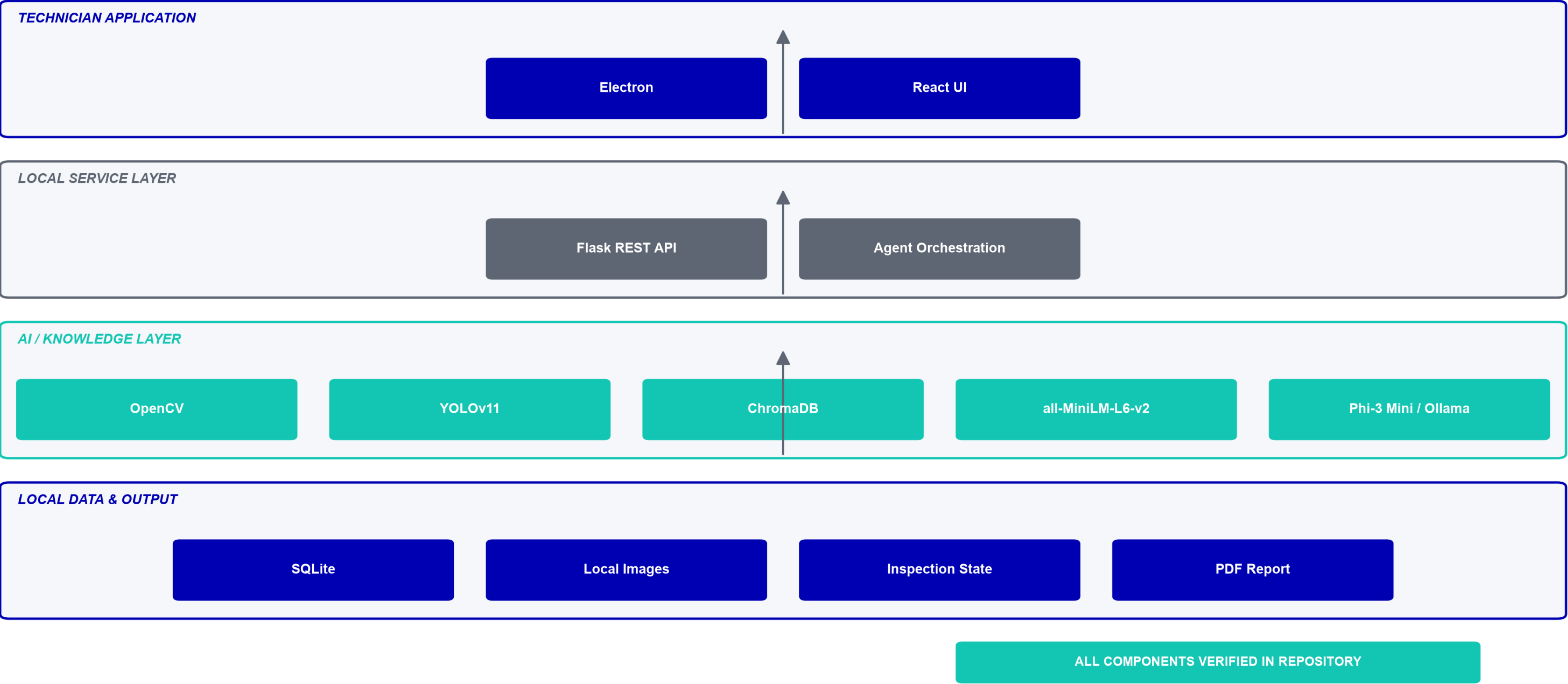
AeroEdge-X AI Inspection Pipeline

CURRENT VERIFIED IMPLEMENTATION

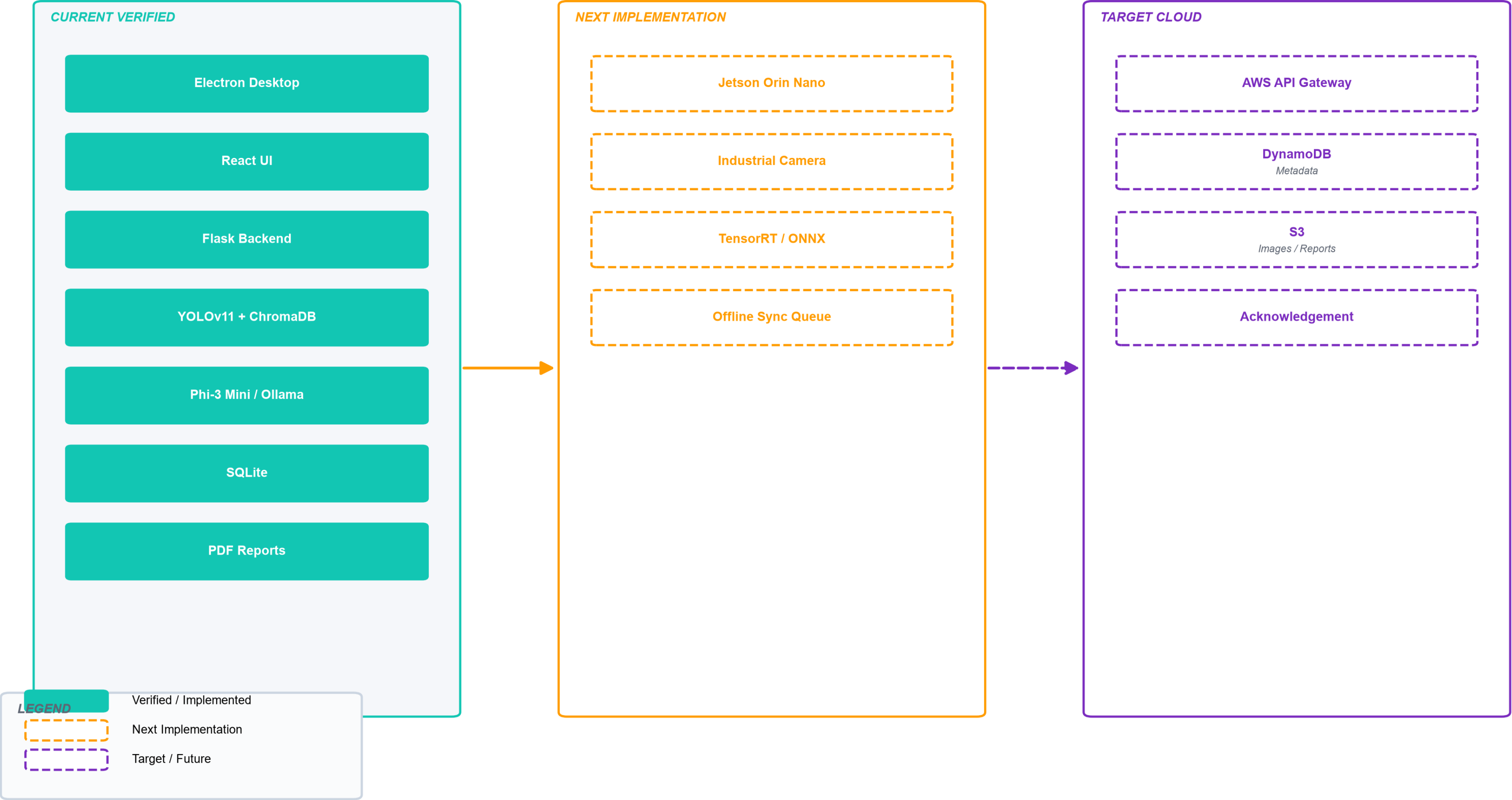


AeroEdge-X Current System Architecture

VERIFIED CURRENT IMPLEMENTATION

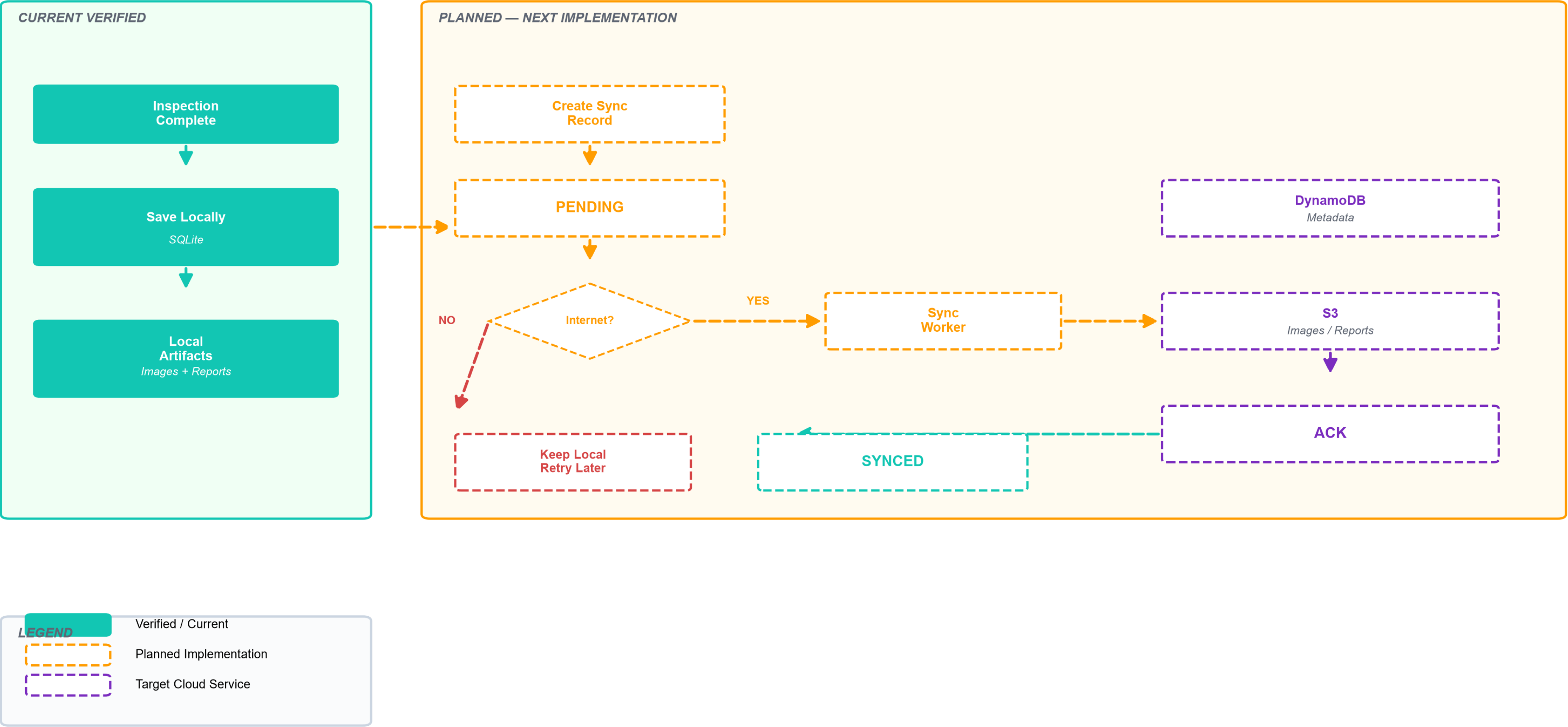


AeroEdge-X Edge + Cloud Evolution



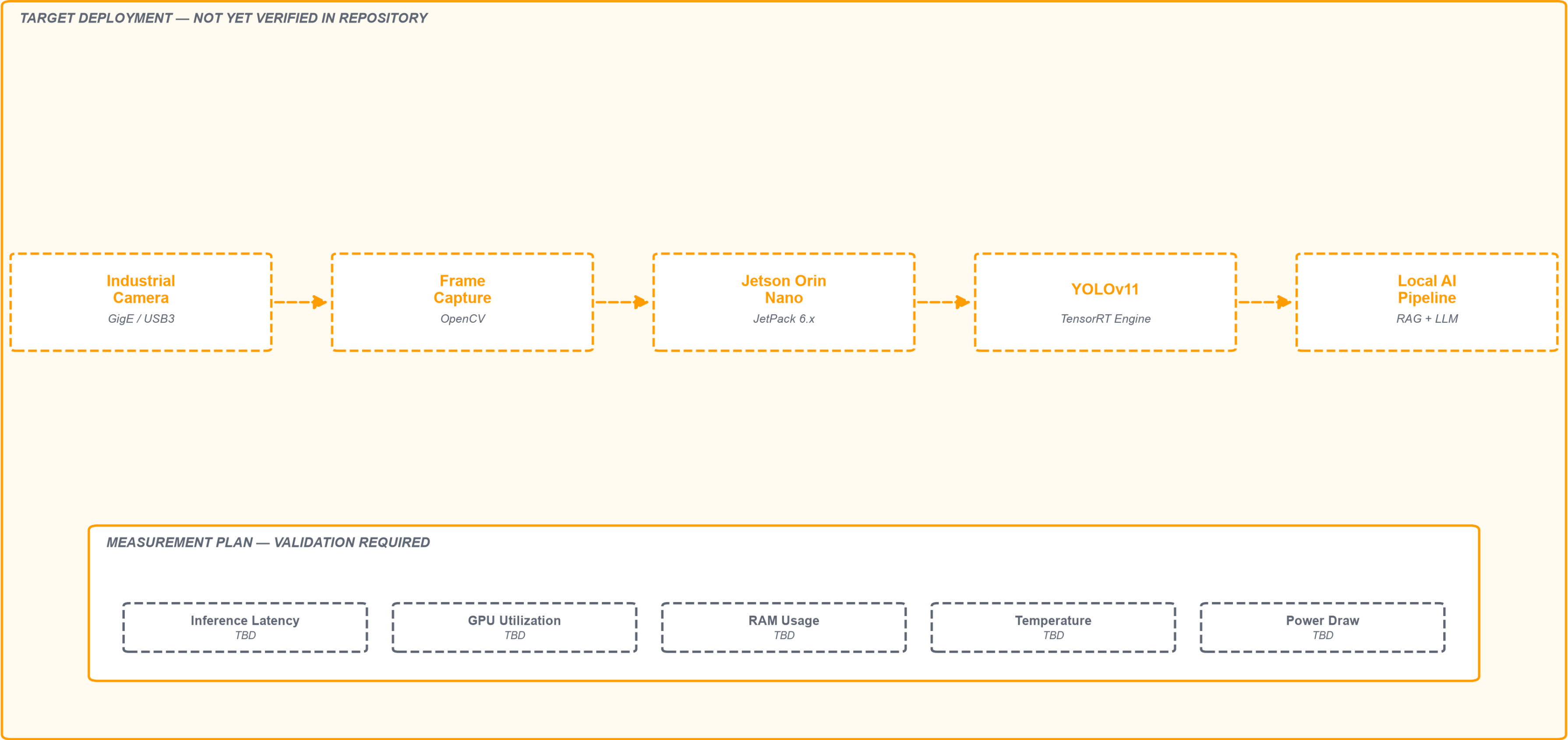
AeroEdge-X Offline-First Synchronization

TARGET / NEXT IMPLEMENTATION



AeroEdge-X Jetson Orin Nano Edge Deployment

TARGET EDGE DEPLOYMENT



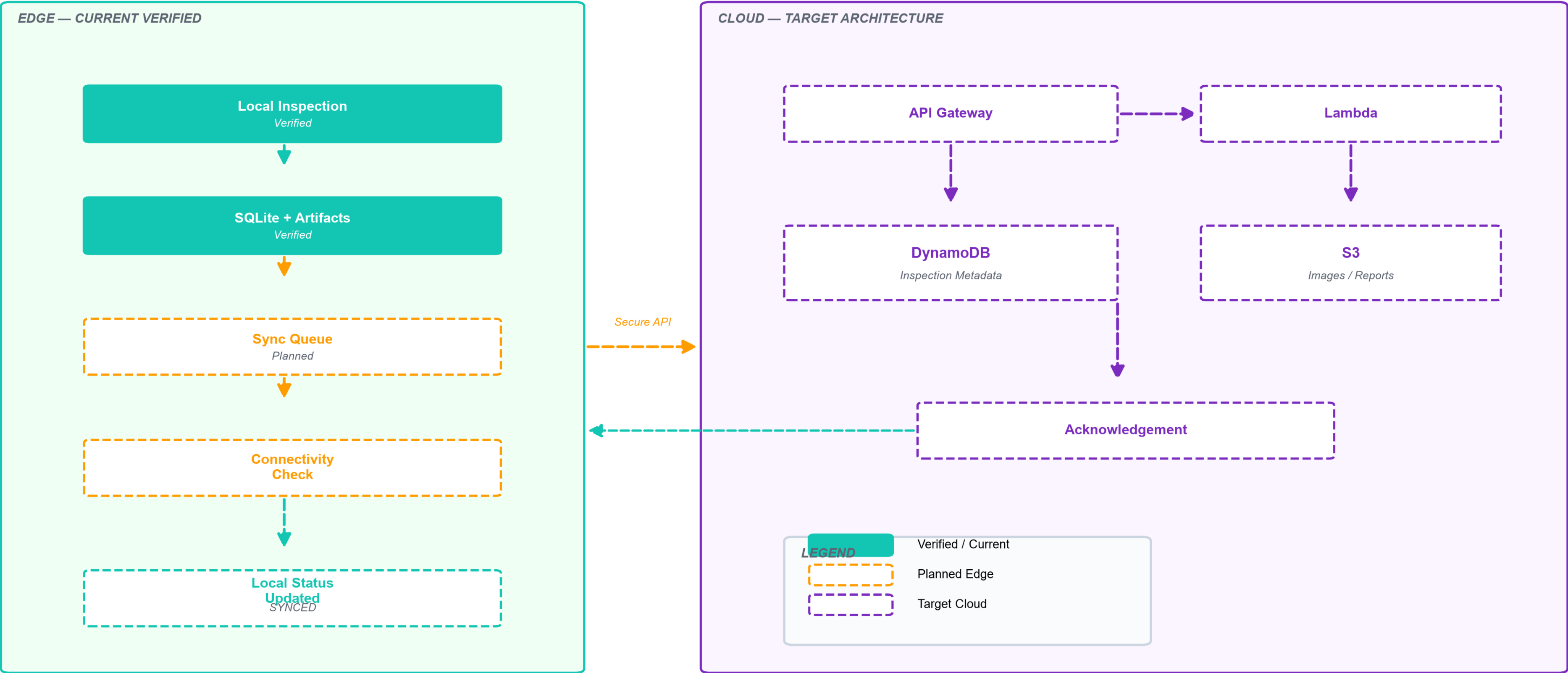
LEGEND

Target / Not Yet Implemented

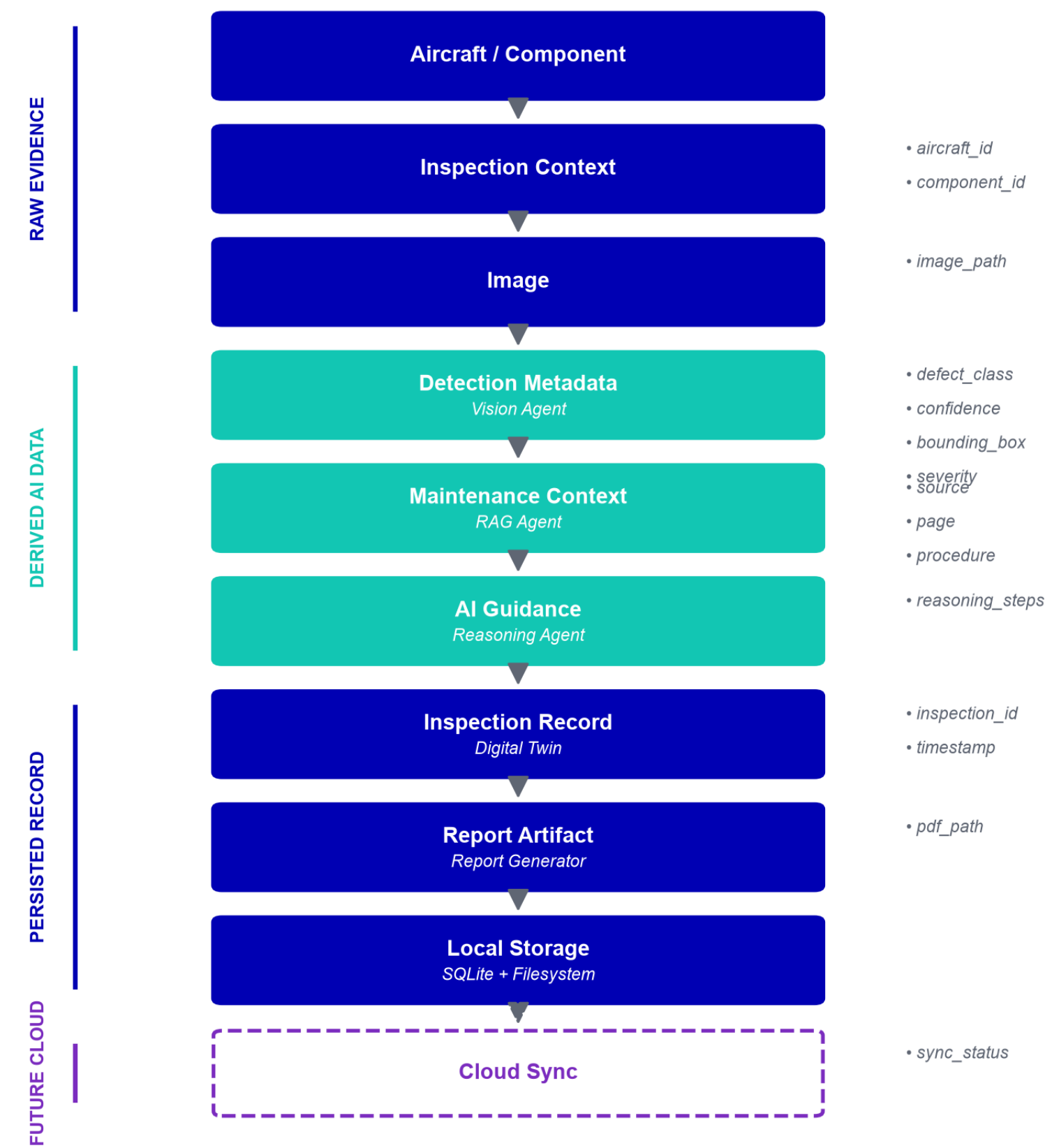
Measurement Pending

AeroEdge-X AWS Synchronization Architecture

TARGET ARCHITECTURE

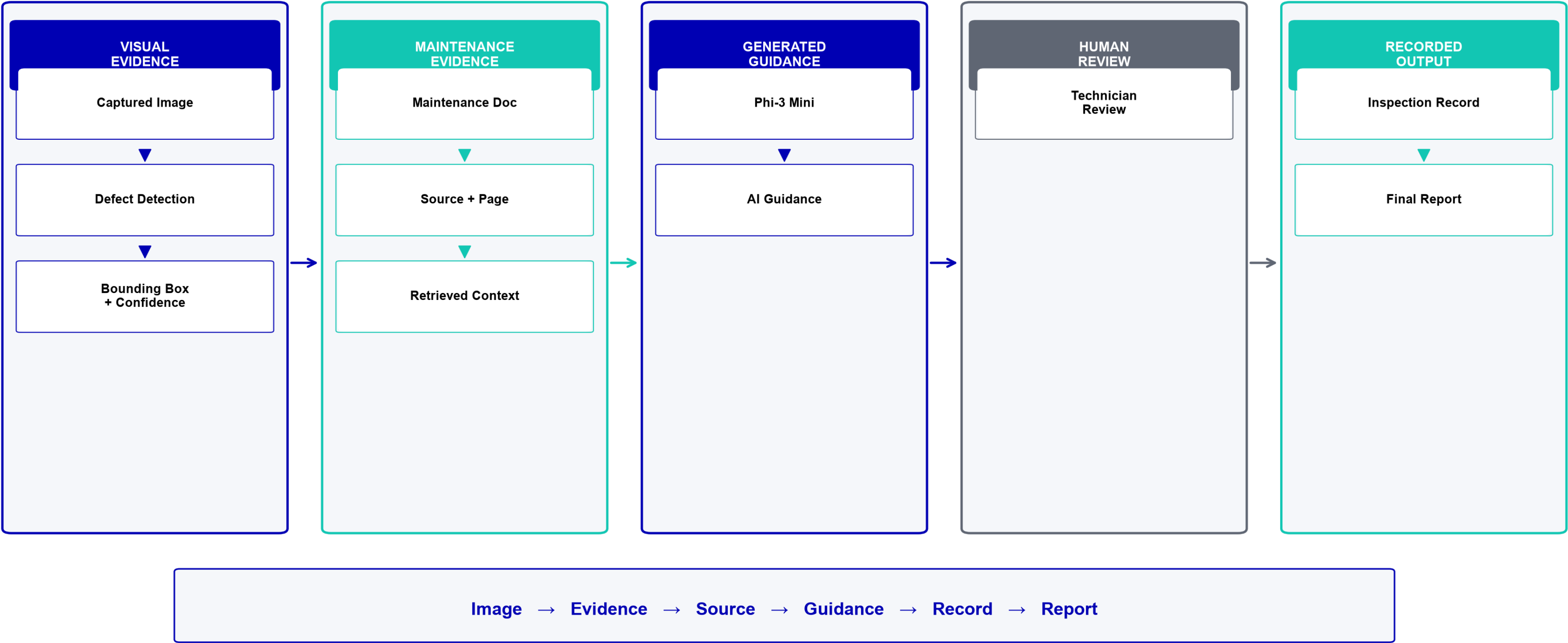


AeroEdge-X Inspection Data Flow

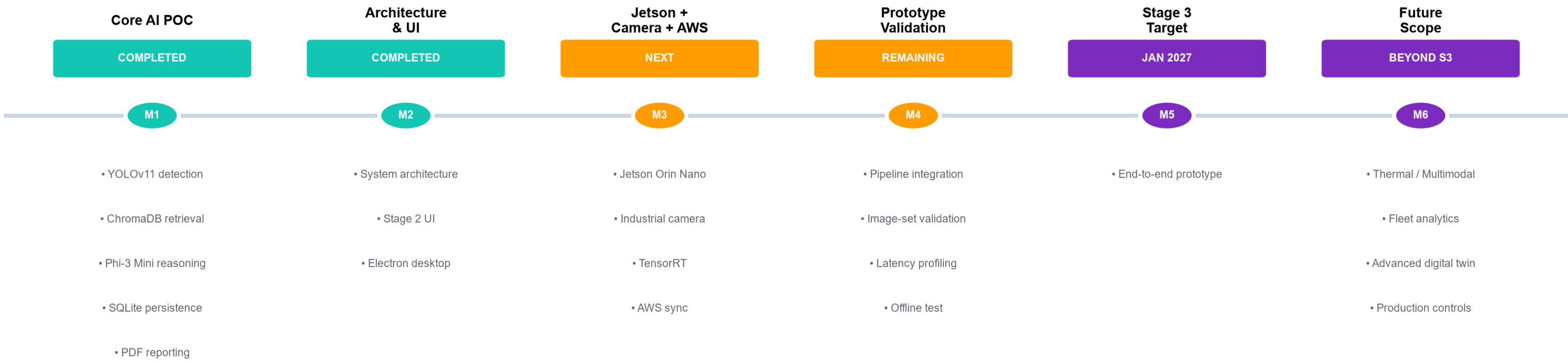


AeroEdge-X Inspection Evidence Traceability

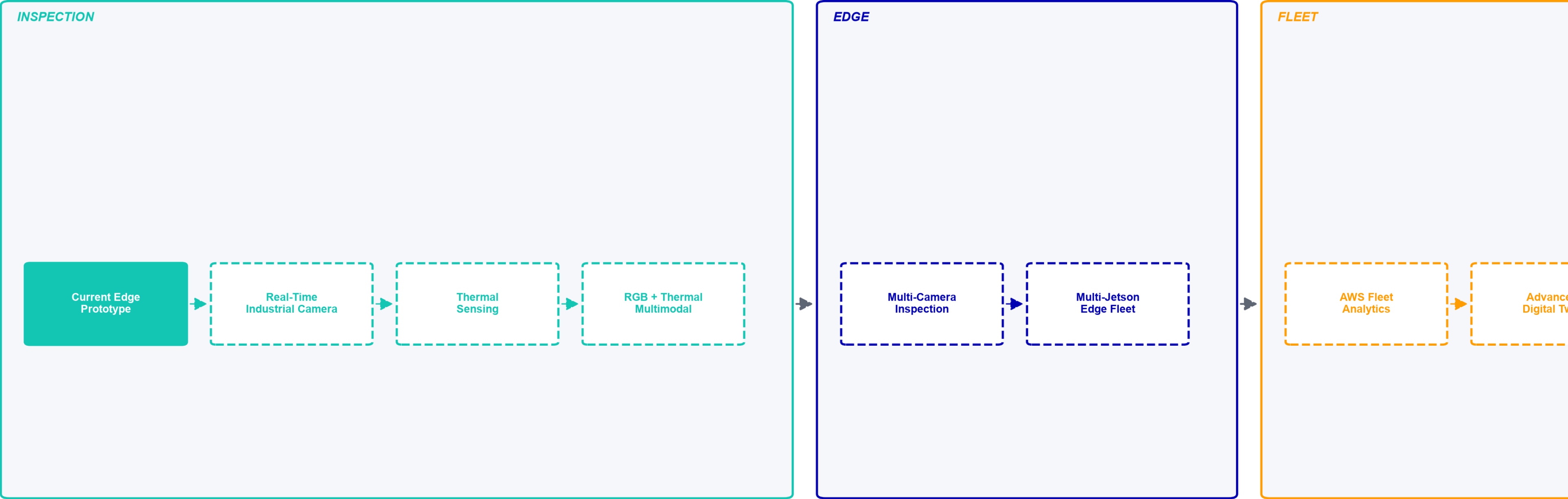
CURRENT VERIFIED CHAIN



AeroEdge-X Current Status & Roadmap



AeroEdge-X Future Evolution



INTELLIG

AeroEdge-X Prototype Validation Plan

Validation Area	Test	Status	Evidence / Measurement
Offline Inspection	Complete workflow without Internet	VERIFIED	rag_agent.py: HF_HUB_OFFLINE=1
Defect Detection	Detection + localization	VERIFIED	vision_agent.py: YOLOv11 via ultralytics
RAG Retrieval	Correct source + page	VERIFIED	rag_agent.py: ChromaDB + PyMuPDFLoader
Local Reasoning	Guidance grounded in retrieved context	VERIFIED	reasoning_agent.py: Ollama Phi-3 Mini
Persistence	Inspection saved locally	VERIFIED	digital_twin.py: SQLite log_inspection()
Reporting	PDF generated	VERIFIED	report_generator.py: ReportLab
Jetson	Edge inference	PLANNED	No JetPack / TensorRT scripts found
Camera	Live image acquisition	PARTIAL	Upload works; no VideoCapture code
Synchronization	Offline queue + reconnect	PLANNED	No sync queue or AWS code found
Recovery	Restart / network failure	PLANNED	Offline queue roadmap
Performance	Latency / RAM / GPU / temperature	PLANNED	Benchmarking planned for Stage 3

AeroEdge-X Architecture Evolution

